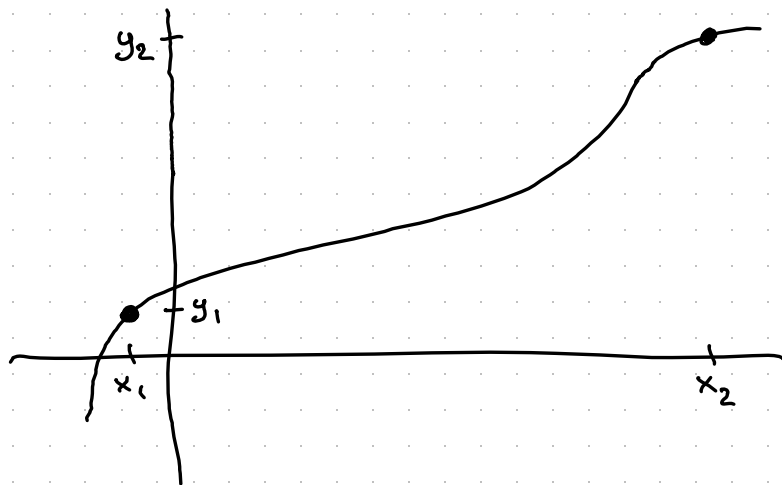


Arc length of parametric curves - 7.2



$$r(t) = (x(t), y(t))$$

$$r(t_1) = (x_1, y_1)$$

$$r(t_2) = (x_2, y_2)$$

$$\int_{t_1}^{t_2} \sqrt{1 + \left(\frac{dy}{dx}\right)^2} \frac{dx}{dt} dt$$

$$\int_{t_1}^{t_2} \sqrt{1 + \left(\frac{dx}{dy}\right)^2} \frac{dy}{dt} dt$$

$$\int_{x_1}^{x_2} \sqrt{1 + \left(\frac{dy}{dx}\right)^2} dx$$

$$\int_{y_1}^{y_2} \sqrt{1 + \left(\frac{dx}{dy}\right)^2} dy$$

$$\int_{t_1}^{t_2} \sqrt{1 + \left(\frac{dy}{dx}\right)^2} \frac{dx}{dt} dt = \int_{t_1}^{t_2} \sqrt{1 + \left(\frac{\frac{dy}{dt}}{\frac{dx}{dt}}\right)^2} \frac{dx}{dt} dt$$

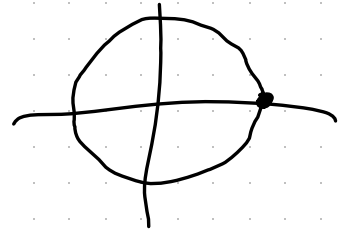
$$= \int_{t_1}^{t_2} \sqrt{\left(1 + \left(\frac{\frac{dy}{dt}}{\frac{dx}{dt}}\right)^2\right) \left(\frac{dx}{dt}\right)^2} dt = \int_{t_1}^{t_2} \sqrt{\left(\frac{dx}{dt}\right)^2 + \left(\frac{dy}{dt}\right)^2} dt$$

$$= \int_{t_1}^{t_2} \sqrt{x'(t)^2 + y'(t)^2} dt$$

$$\int_{t_1}^{t_2} \sqrt{1 + \left(\frac{dx}{dy}\right)^2} \frac{dy}{dt} dt = \dots = \int_{t_1}^{t_2} \sqrt{\left(\frac{dx}{dt}\right)^2 + \left(\frac{dy}{dt}\right)^2} dt$$

Example: What is the circumference of a circle of radius r ?

We parameterize the circle as $(r\cos(t), r\sin(t))$ and we get the full circle once on $0 \leq t \leq 2\pi$.



$$\text{Length is } \int_{t_1}^{t_2} \sqrt{x'(t)^2 + y'(t)^2} dt$$

$$= \int_0^{2\pi} \sqrt{(-r\sin(t))^2 + (r\cos(t))^2} dt = \int_0^{2\pi} \sqrt{r^2\sin^2(t) + r^2\cos^2(t)} dt$$

$$= \int_0^{2\pi} \sqrt{r^2(\sin^2(t) + \cos^2(t))} dt = \int_0^{2\pi} \sqrt{r^2} dt = \int_0^{2\pi} r dt = rt \Big|_0^{2\pi} = 2\pi r.$$